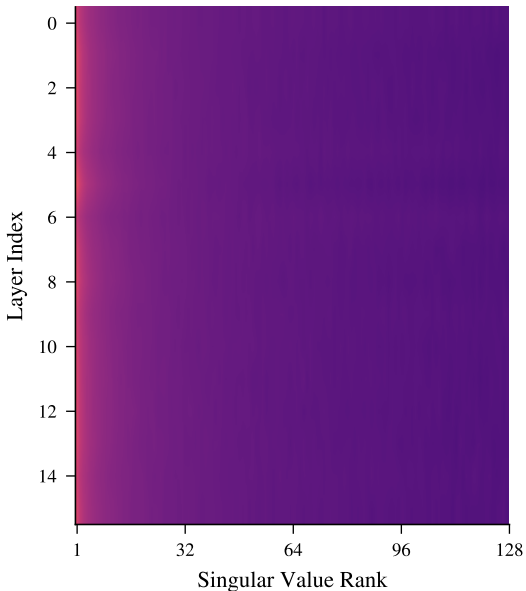


Pythia-1B: Spectral Energy Distribution (simulated from measured α)

Uniform: energy spread across all ranks
(a) Random Init (step 0, $\alpha=4.0$)



Concentrated: energy in top few ranks
(b) Trained (step 143K, $\alpha=2.8$)

